

Sonny Cooper

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PhD Candidate at the University Of Bristol, working as part of the Machine Learning and Computer Vision Group. Researching topics in Video Moment Retrieval tasks.

EDUCATION

- **University of Bristol** 2026-date
PhD Candidate: MaVi Lab
- **University of Bristol** 2022-2026
Computer Science MEng 1st Class Honours
 - **Notable modules:** Individual Project with Innovation Case (73), Applied Deep Learning (73), Advanced Visual AI (69), Image Processing and Computer Vision (71), Machine Learning (77), Data-Driven Computer Science (81), Mathematics for Computer Science B (90), Object Oriented Programming and Algorithms 1 (76), Algorithms 2 (75), Advanced Algorithms (80), Team Project (80)
 - **Masters Dissertation:** Assisting First-Time Buyers in the Housing Market
 - * **Supervisor:** James Cussens
 - * **Scope:** A Comparative Analysis of Machine Learning Models for LSOA Value Forecasts
 - Competed in the 2023 BrisHack 24-hour Hackathon in a team of 5
 - Competed in the unitaryHACK 2025 open source quantum Hackathon in a team of 2
- **Norwich School** 2020-2022
A levels Results: A*A*A*A
 - Computer Science (A*), Maths (A*), Further Maths (A*), Physics (A)
- **Norwich School** 2017-2020
GCSEs Results: 9 subjects ranging 7-9

TECHNICAL SKILLS

- **Languages:** Python, Java, C#, Swift, Go, C
- **Deep Learning and AI:** PyTorch, Hugging Face Transformers, Tensorflow, OpenCV, Scikit-learn, Pandas, NumPy
- **Engineering & Tools:** Git, AWS, CI/CD, Unit Testing

ACADEMIC PROJECT GITHUB

- **Dissertation: Assisting First-Time Buyers in the Housing Market (75)**
 - Compiled an extensive 30-year dataset (1995-2025) consisting of financial data and spatial locality scores (representing schools, universities, business hubs and transportation networks)
 - Constructed three regression models (Multiple Linear Regression, Geographically and Temporally Weighted Regression and XGBoost), alongside a Spacetimeformer model adapted from traffic and weather forecasting
 - Direct classifier pipeline and regress-then-classify pipeline were compared. Using an XGBoost direct classifier two-thirds of all top 25% areas were captured with a precision of 1.6 times that of random guessing
 - XGBoost classifier modified to adapt to various risk levels, achieving a 79% precision in a cautious configuration and 74% recall in an aggressive configuration
- **Comparative Analysis of Deep Learning Methodologies for Single-Image Super-Resolution (75)** (Code and Report)
 - Evaluated GANs (BSRGAN), Diffusion models (SR3), and Transformers (SwinIR) on the DIV2K dataset
 - Demonstrated that Transformer models outperform GANs and Diffusion in noise robustness and LPIPS metrics (0.55 vs 0.64/0.56)
 - Analysed computational efficiency, identifying Diffusion models as computationally expensive (6,020 GFLOPs) compared to Transformers (19.77 GFLOPs)
 - Investigated the impact of various scaling and noise degradation techniques on model performance
- **Siamese ProgressionNet for Temporal Ordering in Egocentric Video (79)** (Code and Report)
 - Developed a Siamese network to determine pairwise temporal ordering in cooking videos using the HD-EPIC dataset

- Conducted studies on data augmentation and learning rates, improving F1 score by 19% through cropping and perspective techniques
- Implemented layer freezing strategies to analyse regularization effects and prevent overfitting on small dataset samples
- **Comparative Analysis of Statistical Learning & Stochastic Models (76)** (Code and Report)
 - Introduced PCA to reduce feature space (54 to 41 dimensions) while retaining 95% variance, and evaluated the efficacy of K-Means and Gaussian Mixture Models (GMM) on high-dimensional, non-independent ecological data
 - Demonstrated the limitations of SVMs on non-linearly separable data and contrasted this with the robustness of Random Forests, which achieved 95.5% accuracy compared to 67% for Logistic Regression
 - Conducted Bayesian Linear Regression by defining priors and sampling from posterior distributions, comparing results against a custom Multi-Layer Perceptron (MLP) to demonstrate the susceptibility of NNs to overfitting on small datasets (MSE 0.23 vs 0.06)

OPEN SOURCE

- **Internalised NomicBERT model for the HuggingFace/Transformers repository** (PR: #43067)
 - Integrated RoPE directly into the self-attention mechanism
 - Replaced standard GELU with SwiGLU to enhance model stability and performance
 - Implemented gradient checkpointing and key-value caching to optimise training memory efficiency and accelerate autoregressive generation
 - Developed weight conversion script to map original Nomic tensors to the Hugging Face architecture
 - Wrote a comprehensive test suite covering unit tests, integration tests, and generation consistency (handling left-padding and attention masks), resolving complex edge cases in batch inference
 - Wrote model documentation describing the model and use cases
- **Introduced qubit routing documentation for the open source quantum circuit optimisation repository PYZX** (PR: #304)

APPLIED ENGINEERING PROJECTS

- **Interactive Museum Exhibit (77)** (Code)
 - Engineered a visually interactive exhibit for Bristol Museum as part of a team of 5, utilising .NET and Avalonia frameworks, alongside implementing unit tests and CI/CD practices
- **End-to-End ANPR Gate System** (Code)
 - Created an ANPR gate system for an industrial park, utilising OpenCV, Matplotlib and Pytesseract to extract number plates and determine access levels for park entry
 - Developed a PHP/MySQL web interface for real-time user management and database logging
- **Real-Time Sepsis Prediction Model**
 - Developed a support tool to predict sepsis from live medical features.
 - Trained a transformer model with a recall of 0.75 and AUC of 0.74, and a LightGBM model with a recall 0.82 and AUC 0.9
- **Computer Vision Stop-Sign Detector** (Code)
 - Implemented a traffic sign detector from scratch without high-level CV libraries, utilising custom implementations of Sobel edge detection and Hough Space transforms to understand fundamental vision algorithms.
- **Distributed Game of Life Simulation (77)**
 - Engineered a high-performance distributed simulation capable of concurrent processing across local cores or AWS instances, implementing parallel execution strategies relevant to large-scale computing.

INTERESTS AND VOLUNTEERING

- **Charity:** Planned and directed a 24-hour walk for Help for Heroes, raising £500 2018
- **C2 World's Longest Tandem Row:** 7 days straight and over 1,100,000m 2020
- **Volunteering:** During A-levels, coached on weekends to teach kids how to row 2021-2022
- **Rowed at Henley Royal Regatta:** Represented both school and university 2021-2023

REFERENCES

- **Dr Steven Ramsay**

- Associate Professor in Programming Languages, University of Bristol
- **Email:** `steven.ramsay@bristol.ac.uk`

- **Dr James Cussens**

- Senior Lecturer, University of Bristol
- **Email:** `james.cussens@bristol.ac.uk`